

Comprehensive Plan: Application Security Session

1. Event Overview

This initiative consists of two independent but strategically connected events:

- Application Security Session (3 Hours)

The objective is to establish a structured pipeline where participants first gain foundational and practical knowledge in application security, and then apply those concepts through project development and presentation.

2. Application Security Session (3 Hours)

2.1 Concept

A focused, hands-on session designed to introduce participants to real-world application vulnerabilities, attacker methodologies, and defensive strategies.

Theme:

“Break it. Understand it. Secure it.”

2.2 Objectives

- Build strong fundamentals in Application Security
- Introduce real-world attack techniques
- Demonstrate exploitation and mitigation
- Bridge theoretical knowledge with practical implementation
- Prepare participants to design and develop security-oriented projects

2.3 Detailed Session Flow (3 Hours)

Hour 1: Foundations

- Introduction to Application Security
- Importance in modern systems (Web applications, APIs, AI-integrated systems)
- Overview of common attack surfaces
- Introduction to widely recognized vulnerability categories
- Core vulnerabilities:
 - SQL Injection
 - Cross-Site Scripting (XSS)
 - Authentication and session management flaws

Outcome: Participants develop a clear understanding of common vulnerabilities and their implications.

Hour 2: Live Demonstration

- Demonstration using a controlled vulnerable application environment
- Step-by-step walkthrough of attack execution:
 - Input manipulation
 - Injection techniques
 - Data extraction methods
- Demonstration of:
 - XSS payload execution
 - Session-related exploitation concepts
- Followed by mitigation techniques:
 - Input validation and sanitization
 - Output encoding
 - Secure configuration practices

Outcome: Participants understand both the attacker's approach and corresponding defensive strategies.

Hour 3: Advanced Concepts

- Secure coding principles
- Fundamentals of secure system and application architecture
- Introduction to emerging security areas:
 - AI system vulnerabilities
 - Prompt injection and misuse scenarios
 - Risks in data-driven and retrieval-based systems
- Discussion of real-world incidents and lessons learned
- Interactive question and answer session

Outcome: Participants gain exposure to advanced, industry-relevant security considerations.

2.4 Requirements

Hardware

- Projector or display system
- Presenter laptop
- Reliable internet connectivity

Software

- Vulnerable application environment (e.g., training lab setup)
- Web browser tools
- Optional: proxy or testing tools for demonstration

Human Resources

- One experienced speaker
- One assistant (optional)

2.5 Engagement Strategy

- Encourage interaction through guided questions
- Emphasize live demonstrations over theoretical slides
- Include short analytical tasks to maintain engagement

2.6 Expected Outcome

Participants will:

- Understand real-world application security risks
- Develop both offensive and defensive thinking approaches
- Be prepared to conceptualize and build security-focused projects